

DBMS – SELLER AND INVENTORY MANAGEMENT

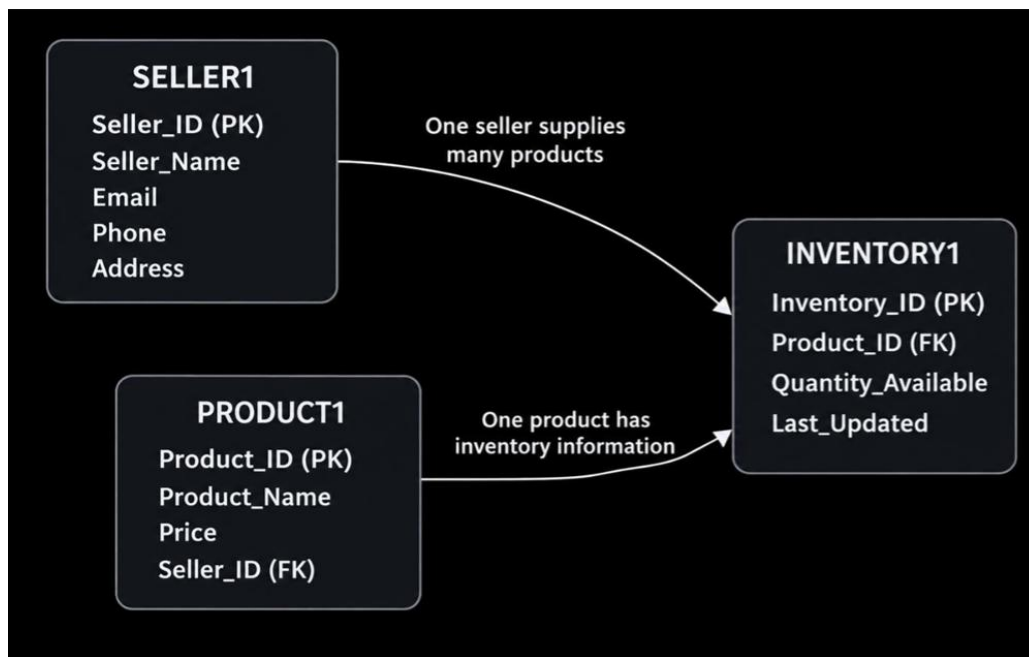
1.Seller

Attribute	Key	Description
Seller_ID	Primary Key (PK)	Unique ID of the seller
Seller_Name	-	Seller's business name
Email	-	Seller's email address
Phone	-	Seller's contact number

2 Inventory

Attribute	Key	Description
Inventory_ID	Primary Key (PK)	Unique inventory ID
Product_ID	Foreign Key (FK)	ID of the product
Quantity_Available	–	Available quantity in stock
Last_Update	–	Date of last stock update

2.2 Relationship



3. Table creation

3.1 Create Seller Table

```
SQL> CREATE TABLE Seller3 (  
2     Seller_ID NUMBER PRIMARY KEY,  
3     Seller_Name VARCHAR2(50),  
4     Email VARCHAR2(100),  
5     Phone VARCHAR2(15)  
6 );
```

Table created.

3.2 Create Inventory Table

```
SQL> CREATE TABLE Inventory4 (  
2     Inventory_ID NUMBER PRIMARY KEY,  
3     Product_ID NUMBER,  
4     Seller_ID NUMBER,  
5     Quantity_Available NUMBER,  
6     Status VARCHAR2(20),  
7     Last_Update DATE,  
8     FOREIGN KEY (Product_ID) REFERENCES Product3(Product_ID),  
9     FOREIGN KEY (Seller_ID) REFERENCES Seller3(Seller_ID)  
10 );
```

Table created.

4. Seller Data Insertion

```
SQL> INSERT INTO Seller3 VALUES (201, 'Music World', 'musicworld@gmail.com', '9876543210');
1 row created.

SQL> INSERT INTO Seller3 VALUES (202, 'Melody Store', 'melodystore@gmail.com', '9876543211');
1 row created.

SQL> INSERT INTO Seller3 VALUES (203, 'Sound Hub', 'soundhub@gmail.com', '9876543212');
1 row created.
```

5. Inventory Data Insertion

```
SQL> INSERT INTO Inventory4
  2 VALUES (401, 101, 201, 45, 'Available', TO_DATE('01-SEP-2026','DD-MON-YYYY'));
1 row created.

SQL>
SQL> INSERT INTO Inventory4
  2 VALUES (402, 102, 202, 40, 'Available', TO_DATE('01-SEP-2026','DD-MON-YYYY'));
1 row created.

SQL>
SQL> INSERT INTO Inventory4
  2 VALUES (403, 103, 203, 0, 'Unavailable', TO_DATE('01-SEP-2026','DD-MON-YYYY'));
1 row created.
```

6. Seller Product Information

6.1 Display Seller and Product Details

```
SQL> SELECT
  2     s.Seller_ID,
  3     s.Seller_Name,
  4     p.Product_ID,
  5     p.Product_Name,
  6     p.Category,
  7     p.Price,
  8     i.Quantity_Available,
  9     i.Status
 10 FROM Seller3 s
 11 JOIN Inventory4 i
 12   ON s.Seller_ID = i.Seller_ID
 13 JOIN Product3 p
 14   ON i.Product_ID = p.Product_ID;
```

SELLER_ID	SELLER_NAME		PRODUCT_ID

PRODUCT_NAME			

CATEGORY		PRICE	QUANTITY_AVAILABLE

STATUS			

201	Music World		101
Yamaha	Guitar		
Guitar		25000	45
Available			
SELLER_ID	SELLER_NAME		PRODUCT_ID

PRODUCT_NAME			

CATEGORY		PRICE	QUANTITY_AVAILABLE

STATUS			

202	Melody Store		102
Casio	Keyboard		
Keyboard		18000	40
Available			
SELLER_ID	SELLER_NAME		PRODUCT_ID

PRODUCT_NAME			

CATEGORY		PRICE	QUANTITY_AVAILABLE

STATUS			

203	Sound Hub		103
Pearl	Drums		
Drums		35000	0
Unavailable			

7. Inventory Status

7.1 Display Available Products

```

SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Quantity_Available,
  6     i.Status
  7 FROM Product3 p
  8 JOIN Inventory4 i
  9     ON p.Product_ID = i.Product_ID
 10 JOIN Seller3 s
 11     ON i.Seller_ID = s.Seller_ID
 12 WHERE i.Status = 'Available';

PRODUCT_ID PRODUCT_NAME
-----
SELLER_NAME QUANTITY_AVAILABLE
-----
STATUS
-----
      101 Yamaha Guitar
Music World 45
Available

      102 Casio Keyboard
Melody Store 40
Available

PRODUCT_ID PRODUCT_NAME
-----
SELLER_NAME QUANTITY_AVAILABLE
-----
STATUS
-----

```

7.2 Display Unavailable Products

```

SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Quantity_Available,
  6     i.Status
  7 FROM Product3 p
  8 JOIN Inventory4 i
  9     ON p.Product_ID = i.Product_ID
 10 JOIN Seller3 s
 11     ON i.Seller_ID = s.Seller_ID
 12 WHERE i.Status = 'Unavailable';

PRODUCT_ID PRODUCT_NAME
-----
SELLER_NAME QUANTITY_AVAILABLE
-----
STATUS
-----
      103 Pearl Drums
Sound Hub 0
Unavailable

```

8. Inventory Update Demonstrate updating stock information

```
SQL> SELECT
  2     i.Inventory_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Quantity_Available,
  6     i.Status,
  7     i.Last_Update
  8 FROM Inventory4 i
  9 JOIN Product3 p
10     ON i.Product_ID = p.Product_ID
11 JOIN Seller3 s
12     ON i.Seller_ID = s.Seller_ID
13 ORDER BY i.Inventory_ID;
```

INVENTORY_ID PRODUCT_NAME

SELLER_NAME

QUANTITY_AVAILABLE

STATUS

LAST_UPDA

401 Yamaha Guitar	
Music World	45
Available	01-SEP-26

402 Casio Keyboard	
Melody Store	40
Available	01-SEP-26

INVENTORY_ID PRODUCT_NAME

SELLER_NAME

QUANTITY_AVAILABLE

STATUS

LAST_UPDA

403 Pearl Drums	
Sound Hub	0
Unavailable	01-SEP-26

9. Inventory Status Reports

9.1 Complete Inventory Report

```
2      i.Inventory_ID,  
3      s.Seller_Name,  
4      p.Product_Name,  
5      p.Price,  
6      i.Quantity_Available,  
7      CASE  
8          WHEN i.Quantity_Available > 0 THEN 'Available'  
9          ELSE 'Unavailable'  
10     END AS Stock_Status,  
11     i.Last_Update  
12 FROM INVENTORY4 i  
13 JOIN PRODUCT3 p  
14     ON i.Product_ID = p.Product_ID  
15 JOIN SELLER3 s  
16     ON i.Seller_ID = s.Seller_ID  
17 ORDER BY i.Inventory_ID;
```

INVENTORY_ID SELLER_NAME

PRODUCT_NAME PRICE QUANTITY_AVAILABLE

STOCK_STATU LAST_UPDA

401 Music World
Yamaha Guitar 25000 45
Available 01-SEP-26

402 Melody Store
Casio Keyboard 18000 40
Available 01-SEP-26

INVENTORY_ID SELLER_NAME

PRODUCT_NAME PRICE QUANTITY_AVAILABLE

STOCK_STATU LAST_UPDA

403 Sound Hub
Pearl Drums 35000 0
Unavailable 01-SEP-26

9.2 Available and Unavailable Product Count

```
SQL> SELECT
  2     CASE
  3         WHEN Quantity_Available > 0 THEN 'Available'
  4         ELSE 'Unavailable'
  5     END AS Stock_Status,
  6     COUNT(*) AS Total_Products
  7 FROM INVENTORY4
  8 GROUP BY
  9     CASE
 10         WHEN Quantity_Available > 0 THEN 'Available'
 11         ELSE 'Unavailable'
 12     END;

STOCK_STATU TOTAL_PRODUCTS
-----
Available          2
Unavailable         1
```

9.3 Total Stock by Seller

```
SQL> SELECT
  2     s.Seller_Name,
  3     SUM(i.Quantity_Available) AS Total_Stock
  4 FROM SELLER3 s
  5 JOIN INVENTORY4 i
  6     ON s.Seller_ID = i.Seller_ID
  7 JOIN PRODUCT3 p
  8     ON i.Product_ID = p.Product_ID
  9 GROUP BY s.Seller_Name
 10 ORDER BY s.Seller_Name;

SELLER_NAME                                     TOTAL_STOCK
-----
Melody Store                                     40
Music World                                     45
Sound Hub                                       0
```

9.4 Products with Zero Stock

This is particularly useful for demonstrating unavailable products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Quantity_Available,
  6     CASE
  7         WHEN i.Quantity_Available = 0 THEN 'Unavailable'
  8         ELSE 'Available'
  9     END AS Stock_Status
 10 FROM PRODUCT3 p
 11 JOIN INVENTORY4 i
 12     ON p.Product_ID = i.Product_ID
 13 JOIN SELLER3 s
 14     ON i.Seller_ID = s.Seller_ID
 15 WHERE i.Quantity_Available = 0;
```

PRODUCT_ID	PRODUCT_NAME	SELLER_NAME	QUANTITY_AVAILABLE	STOCK_STATU
103	Pearl Drums	Sound Hub	0	Unavailable